

Homework 7

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*All of my homework is available at <https://github.com/lacampanella233/Homework.git>.

Problem 1 (5C-1). Prove or give a counterexample: If $T \in \mathcal{L}(V)$ and T^2 has an upper-triangular matrix with respect to some basis of V , then T has an upper-triangular matrix with respect to some basis of V .

Solution. We give a counterexample. Let $V = \mathbb{R}^2$ and let $T \in \mathcal{L}(V)$ be defined by rotating every vector in V by 90° counterclockwise. Then $T^2 = -I$, which obviously has an upper-triangular matrix. However, T does not have an upper-triangular matrix with respect to any basis of V since it has no eigenvalues in $\mathbb{R}$.

Problem 2 (5C-2). Suppose A and B are upper-triangular matrices of the same size, with $\alpha_1, \dots, \alpha_n$ on the diagonal of A and $\beta_1, \dots, \beta_n$ on the diagonal of B .

- (a) Show that $A + B$ is an upper-triangular matrix with $\alpha_1 + \beta_1, \dots, \alpha_n + \beta_n$ on the diagonal.
- (b) Show that AB is an upper-triangular matrix with $\alpha_1\beta_1, \dots, \alpha_n\beta_n$ on the diagonal.

Proof. (a) Denote M_{ij} as the (i, j) -th entry of a matrix M . Since A and B are upper-triangular, we have $A_{ij} = 0$ and $B_{ij} = 0$ for all $i > j$. Therefore, for all $i > j$ and k , we have

$$(A + B)_{ij} = A_{ij} + B_{ij} = 0 + 0 = 0, \quad (A + B)_{kk} = A_{kk} + B_{kk} = \alpha_k + \beta_k.$$

(b) For all $i > j$, we have

$$(AB)_{ij} = \sum_{k=1}^n A_{ik}B_{kj} = 0,$$

since for each term in the sum, either $i > k$ or $k > j$. For the diagonal entries, we have

$$(AB)_{kk} = \sum_{j=1}^n A_{kj}B_{jk} = A_{kk}B_{kk} = \alpha_k\beta_k.$$

□

Problem 3 (5C-3). Suppose $T \in \mathcal{L}(V)$ is invertible and $v_1, \dots, v_n$ is a basis of V with respect to which the matrix of T is upper triangular, with $\lambda_1, \dots, \lambda_n$ on the diagonal. Show that the matrix of T^{-1} is also upper triangular with respect to the basis $v_1, \dots, v_n$, with

$$\frac{1}{\lambda_1}, \dots, \frac{1}{\lambda_n}$$

on the diagonal.

Proof. Denote the diagonal entries of the matrix of T^{-1} by $\mu_1, \dots, \mu_n$. By the previous problem, the matrix of $TT^{-1} = I$ is upper-triangular with $\lambda_1\mu_1, \dots, \lambda_n\mu_n$ on the diagonal. Since the matrix of I has 1 on the diagonal, we have $\lambda_k\mu_k = 1$ for all $k = 1, \dots, n$. Therefore, $\mu_k = 1/\lambda_k$ for all $k = 1, \dots, n$. $\square$

Problem 4 (5C-8). Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$, and there exists a nonzero vector $v \in V$ such that $T^2v + 2Tv = -2v$.

- (a) Prove that if $\mathbb{F} = \mathbb{R}$, then there does not exist a basis of V with respect to which T has an upper-triangular matrix.
- (b) Prove that if $\mathbb{F} = \mathbb{C}$ and A is an upper-triangular matrix that equals the matrix of T with respect to some basis of V , then $-1 + i$ or $-1 - i$ appears on the diagonal of A .

Proof. (a) Denote the matrix of T with respect to some basis of V by A , such that A is upper triangular. Then the matrix of $T^2 + 2T + 2I$ is $A^2 + 2A + 2I$, which is also upper triangular and with diagonal entries $\lambda^2 + 2\lambda + 2$, where λ is a diagonal entry of A . Since $\lambda \in \mathbb{R}$, we have $\lambda^2 + 2\lambda + 2 > 0$, which shows that $A^2 + 2A + 2I$ is invertible. However, since $T^2v + 2Tv = -2v$, we have $(T^2 + 2T + 2I)v = 0$, which contradicts the invertibility of $A^2 + 2A + 2I$.

(b) The same argument as in (a) shows that $A^2 + 2A + 2I$ is not invertible, which means that there exists a diagonal entry λ of A such that $\lambda^2 + 2\lambda + 2 = 0$. Solving this equation gives $\lambda = -1 + i$ or $\lambda = -1 - i$. $\square$

Problem 5 (5C-12). Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$ has an upper-triangular matrix with respect to some basis of V , and U is a subspace of V that is invariant under T .

- (a) Prove that $T|_U$ has an upper-triangular matrix with respect to some basis of U .
- (b) Prove that the quotient operator T/U has an upper-triangular matrix with respect to some basis of V/U .

Proof. (a) Let $v_1, \dots, v_n$ be a basis of V such that the matrix of T with respect to this basis is upper triangular, and let $V_k = \text{span}\{v_1, \dots, v_k\}$. Then, $\dim V_k = k$, and $\{0\} = V_0 \subset V_1 \subset \dots \subset V_n = V$ is a sequence of subspaces that are invariant under T .

We will show that $\{0\} = (V_0 \cap U) \subset (V_1 \cap U) \subset \dots \subset (V_n \cap U) = U$ is a sequence of subspaces that are invariant under $T|_U$ with $\dim(V_{k+1} \cap U) - \dim(V_k \cap U) \leq 1$ for all $k = 0, \dots, n-1$. Since

$V_{k+1} = V_k + \text{span}\{v_{k+1}\}$, we have

$$V_{k+1} \cap U = (V_k + \text{span}\{v_{k+1}\}) \cap U = (V_k \cap U) + (\text{span}\{v_{k+1}\} \cap U).$$

Therefore

$$\dim(V_{k+1} \cap U) = \dim(V_k \cap U) + \dim(\text{span}\{v_{k+1}\} \cap U) - \dim(V_k \cap \text{span}\{v_{k+1}\} \cap U) \leq \dim(V_k \cap U) + 1.$$

Denote $m = \dim U$. Then there exists $0 = i_0 < i_1 < \cdots < i_m = n$ such that $\dim(V_{i_k} \cap U) = k$ for all $k = 0, \dots, m$. Pick $u_k \in (V_{i_k} \cap U) - (V_{i_{k-1}} \cap U)$, then $\{u_1, \dots, u_m\}$ is a basis of U . Moreover, for any $1 \leq k \leq m$, $\{u_1, \dots, u_k\}$ is a basis of $V_{i_k} \cap U$. Therefore, the matrix of $T|_U$ with respect to the basis $u_1, \dots, u_m$ is upper triangular.

(b) We will consider the sequence of subspaces

$$\{0\} = (V_0 + U)/U \subset (V_1 + U)/U \subset \cdots \subset (V_n + U)/U = V/U.$$

Note that

$$\begin{aligned} \dim((V_{k+1} + U)/U) &= \dim(V_{k+1} + U) - \dim U = \dim V_{k+1} + \dim U - \dim(V_{k+1} \cap U) - \dim U \\ &= \dim V_{k+1} - \dim(V_{k+1} \cap U) \leq 1 + \dim V_k - \dim(V_k \cap U) \\ &= 1 + \dim((V_k + U)/U). \end{aligned}$$

Then the similar argument as in (a) shows that there exists a basis of V/U with respect to which the matrix of T/U is upper triangular. $\square$

Problem 6 (5C-13). Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Suppose there exists a subspace U of V that is invariant under T such that $T|_U$ has an upper-triangular matrix with respect to some basis of U and also T/U has an upper-triangular matrix with respect to some basis of V/U . Prove that T has an upper-triangular matrix with respect to some basis of V .

Proof. Denote $\{u_1, \dots, u_m\}$ the basis of U such that the matrix of $T|_U$ with respect to this basis is upper triangular, and denote $\{v_1 + U, \dots, v_n + U\}$ the basis of V/U such that the matrix of T/U with respect to this basis is upper triangular. We will show that $\{u_1, \dots, u_m, v_1, \dots, v_n\}$ is a basis of V such that the matrix of T with respect to this basis is upper triangular.

First, for any $1 \leq k \leq m$, we have $\text{span}\{u_1, \dots, u_k\}$ is T -invariant since the matrix of $T|_U$ with respect to the basis $u_1, \dots, u_m$ is upper triangular. Next, for any $1 \leq k \leq n$, we have

$$\begin{aligned} T(v_k) + U &= (T/U)(v_k + U) \in \text{span}\{v_1 + U, \dots, v_k + U\} \subset \text{span}\{u_1, \dots, u_m, v_1, \dots, v_k\} + U \\ &\implies T(v_k) \in \text{span}\{u_1, \dots, u_m, v_1, \dots, v_k\}. \end{aligned}$$

Then, $\text{span}\{u_1, \dots, u_m, v_1, \dots, v_k\}$ is T -invariant for all $1 \leq k \leq n$. Therefore, the matrix of T with respect to the basis $u_1, \dots, u_m, v_1, \dots, v_n$ is upper triangular. $\square$

Problem 7 (5C-14). Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that T has an upper-triangular matrix with respect to some basis of V if and only if the dual operator T^* has an upper-triangular matrix with respect to some basis of the dual space V^* .

Proof. Since V is finite-dimensional, we can treat V^{**} as V . Then we only need to show “ $\implies$ ”.

Let $\mathcal{B} := \{v_1, \dots, v_n\}$ be a basis of V such that the matrix of T with respect to this basis is upper-triangular, and let $\mathcal{B}^* := \{\varphi_1, \dots, \varphi_n\}$ be the dual basis of V^* . Then we have

$$\mathcal{M}(T, \mathcal{B}) = (\mathcal{M}(T^*, \mathcal{B}^*))^T.$$

Then the matrix of T^* with respect to the basis $\mathcal{B}^*$ is lower-triangular matrix, which implies that the matrix of T^* with respect to the basis $\{\varphi_n, \varphi_{n-1}, \dots, \varphi_1\}$ is upper triangular. $\square$

Problem 8 (5D-1). Suppose V is a finite-dimensional complex vector space and $T \in \mathcal{L}(V)$.

- (a) Prove that if $T^4 = I$, then T is diagonalizable.
- (b) Prove that if $T^4 = T$, then T is diagonalizable.
- (c) Give an example of an operator $T \in \mathcal{L}(\mathbb{C}^2)$ such that $T^4 = T^2$ and T is not diagonalizable.

Proof. (a) We have

$$(T - I)(T + I)(T - iI)(T + iI) = 0.$$

Since the four polynomials $z - 1, z + 1, z - i, z + i$ are pairwise relatively prime, we have $V = E(1, T) \oplus E(-1, T) \oplus E(i, T) \oplus E(-i, T)$. Therefore, T is diagonalizable.

(b) Similar to (a), we have

$$V = E(0, T) \oplus E(1, T) \oplus E(\omega, T) \oplus E(\omega^2, T), \text{ where } \omega = e^{2\pi i/3}.$$

Therefore, T is diagonalizable.

(c) Let $T \in \mathcal{L}(\mathbb{C}^2)$ be defined by $T(z, w) = (0, z)$ for all $(z, w) \in \mathbb{C}^2$. Then $T^2(z, w) = T(0, z) = (0, 0)$, so $T^2 = 0$, but T is not diagonalizable since T has only one eigenvalue 0 but the eigenspace $E(0, T)$ is one-dimensional. $\square$

Problem 9 (5D-3). Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that if the operator T is diagonalizable, then

$$V = \ker T \oplus \operatorname{im} T.$$

Proof. Let $\{v_1, v_2, \dots, v_n\}$ be all the nonzero eigenvectors of T such that $Tv_k = \lambda_k v_k$ for all $k = 1, \dots, n$. Then $\{v_1, v_2, \dots, v_n\}$ is a basis of V . It's clear that

$$\ker T = \operatorname{span}\{v_i : \lambda_i = 0\} \text{ and } \operatorname{im} T = \operatorname{span}\{v_i : \lambda_i \neq 0\}.$$

Then $V = \ker T \oplus \operatorname{im} T$. $\square$

Problem 10 (5D-5). Suppose V is a finite-dimensional complex vector space and $T \in \mathcal{L}(V)$. Prove that T is diagonalizable if and only if

$$V = \ker(T - \lambda I) \oplus \operatorname{im}(T - \lambda I)$$

for every $\lambda \in \mathbb{C}$.

Proof. “ $\implies$ ”: Let $\{\lambda_i\}$ be all the eigenvalues of T and $\{v_i\}$ be the corresponding eigenvectors. Then $\{v_i\}$ is a basis of V . For any $\lambda \in \mathbb{C}$, similar to the previous problem, we have

$$\ker(T - \lambda I) = \operatorname{span}\{v_i : \lambda_i = \lambda\} \text{ and } \operatorname{im}(T - \lambda I) = \operatorname{span}\{v_i : \lambda_i \neq \lambda\}.$$

Therefore, $V = \ker(T - \lambda I) \oplus \operatorname{im}(T - \lambda I)$.

“ $\impliedby$ ”: We will prove first that

$$\ker(T - \lambda I)^k = \ker(T - \lambda I) \text{ for any } k \geq 1.$$

We will use induction on k . The case $k = 1$ is trivial. Assume that the statement holds for some $k \geq 1$. For any $v \in \ker(T - \lambda I)^{k+1}$, we have $(T - \lambda I)v \in \ker(T - \lambda I)^k = \ker(T - \lambda I)$. Therefore, $(T - \lambda I)v \in \ker(T - \lambda I) \cap \operatorname{im}(T - \lambda I) = \{0\}$, which implies that $v \in \ker(T - \lambda I)$. Therefore, $\ker(T - \lambda I)^{k+1} = \ker(T - \lambda I)$.

Denote all distinct eigenvalues of T by $\lambda_1, \dots, \lambda_m$. Denote the minimal polynomial by $m(\lambda) = (\lambda - \lambda_1)^{d_1} \dots (\lambda - \lambda_m)^{d_m}$. Then

$$V = \bigoplus_{k=1}^m \ker(T - \lambda_k I)^{d_k} = \bigoplus_{k=1}^m \ker(T - \lambda_k I).$$

Therefore, T is diagonalizable. □

Problem 11 (5D-11). Find $T \in \mathcal{L}(\mathbb{C}^3)$ such that 6 and 7 are eigenvalues of T and such that T does not have a diagonal matrix with respect to any basis of $\mathbb{C}^3$.

Solution. Take $T(z_1, z_2, z_3) = (7z_1, 6z_2 + z_3, 6z_3)$. Then T has eigenvalues 6 and 7, but T is not diagonalizable since the eigenspace $E(6, T)$ is one-dimensional.

Problem 12 (5D-16). Suppose that $T \in \mathcal{L}(V)$ is diagonalizable. Let $\lambda_1, \dots, \lambda_m$ denote the distinct eigenvalues of T . Prove that a subspace U of V is invariant under T if and only if there exist subspaces $U_1, \dots, U_m$ of V such that $U_k \subseteq E(\lambda_k, T)$ for each k and

$$U = U_1 \oplus \dots \oplus U_m.$$

Proof. “ $\impliedby$ ”: Since $E(\lambda_k, T)$ is invariant under T for each k , the subspace $U_k \subseteq E(\lambda_k, T)$ is also invariant under T . Therefore, $U = U_1 \oplus \dots \oplus U_m$ is invariant under T .

“ $\implies$ ”: Let $U_k = U \cap E(\lambda_k, T)$, we will show that $U = U_1 \oplus \cdots \oplus U_m$. Since $\bigoplus_{k=1}^m E(\lambda_k, T) = V$ is direct sum, we have $\bigoplus_{k=1}^m U_k$ is also direct sum. For any $u \in U$, we can write $u = u_1 + \cdots + u_m$ with $u_k \in E(\lambda_k, T)$ for all $k = 1, \dots, m$. Since U is invariant under T , we have

$$u_k = \prod_{i \neq k} \frac{(T - \lambda_i I)}{\lambda_k - \lambda_i}(u) \in U \implies u_k \in U \cap E(\lambda_k, T) = U_k$$

for all $k = 1, \dots, m$. Therefore, $U_1 \oplus \cdots \oplus U_m \subset U$. On the other hand, for any $u_k \in U_k$, we have $u_k \in U$ and $u_1 + \cdots + u_m \in U$. Therefore, $U = U_1 \oplus \cdots \oplus U_m$. $\square$

Problem 13 (5D-18). Suppose that $T \in \mathcal{L}(V)$ is diagonalizable and U is a subspace of V that is invariant under T . Prove that the quotient operator T/U is a diagonalizable operator on V/U .

Proof. Denote the distinct eigenvectors of T by $v_1, \dots, v_n$ with $Tv_k = \lambda_k v_k$. Then we have

$$(T/U)(v_k + U) = Tv_k + U = \lambda_k v_k + U = \lambda_k(v_k + U)$$

for all $k = 1, \dots, n$. Then $v_k + U$ is an eigenvector of T/U corresponding to the eigenvalue λ_k .

Note that $\text{span}\{v_k + U : 1 \leq k \leq n\} = \text{span}\{v_k : 1 \leq k \leq n\} + U = V/U$, so we can pick a subset of $\{v_k + U : 1 \leq k \leq n\}$ as a basis of V/U . Therefore, T/U is diagonalizable. $\square$

Problem 14 (5D-19). Prove or give a counterexample: If $T \in \mathcal{L}(V)$ and there exists a subspace U of V that is invariant under T such that $T|_U$ and T/U are both diagonalizable, then T is diagonalizable.

Proof. We give a counterexample. Let $T \in \mathcal{L}(\mathbb{C}^2)$ be defined by $T(z, w) = (z + w, w)$ for all $(z, w) \in \mathbb{C}^2$. Then $T|_{\text{span}\{(0,1)\}}$ is diagonalizable and $T/\text{span}\{(0,1)\}$ is also diagonalizable, but T is not diagonalizable since the eigenspace $E(1, T)$ is one-dimensional. $\square$

Problem 15 (5E-4). Prove or give a counterexample: If A is a diagonal matrix and B is an upper-triangular matrix of the same size as A , then A and B commute.

Solution. We give a counterexample. Take

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Then

$$AB = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \neq \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix} = BA.$$

Problem 16 (5E-7). Suppose V is a complex vector space, $S \in \mathcal{L}(V)$ is diagonalizable, and $T \in \mathcal{L}(V)$ commutes with S . Prove that there is a basis of V such that S has a diagonal matrix with respect to this basis and T has an upper-triangular matrix with respect to this basis.

Proof. Let $\{\lambda_i\}$ be all the distinct eigenvalues of S . Then for any λ_k and any $v \in E(\lambda_k, S)$, we have

$$S(Tv) = T(Sv) = T(\lambda_k v) = \lambda_k(Tv),$$

which implies that $Tv \in E(\lambda_k, S)$. Therefore, $E(\lambda_k, S)$ is invariant under T for all k . Then we can pick a basis of $E(\lambda_k, S)$ with respect to which the matrix of $T|_{E(\lambda_k, S)}$ is upper triangular for all k .

Then the union of these bases is a basis of V , with respect to which the matrix of S is diagonal and the matrix of T is upper triangular. Moreover, since $V = \bigoplus_{k=1}^n E(\lambda_k, S)$, we can infer that $\mathcal{M}(T)$ is block diagonal with blocks $\mathcal{M}(T|_{E(\lambda_k, S)})$ for $k = 1, \dots, n$, which is upper triangular. $\square$

Problem 17 (5E-9). Suppose V is a finite-dimensional nonzero complex vector space. Suppose that $\mathcal{E} \subseteq \mathcal{L}(V)$ is such that S and T commute for all $S, T \in \mathcal{E}$.

- (a) Prove that there is a vector in V that is an eigenvector for every element of $\mathcal{E}$.
- (b) Prove that there is a basis of V with respect to which every element of $\mathcal{E}$ has an upper-triangular matrix.

Proof. (a) We will use induction on $\dim V$. The case $\dim V = 1$ is trivial. Assume that the statement holds for any complex vector space with dimension less than n for some $n \geq 2$.

Let $S \in \mathcal{E}$ be a non-scalar operator, then there exists an eigenvalue λ of S such that the eigenspace $E(\lambda, S)$ is a nonzero proper subspace of V . Since S commutes with every element of $\mathcal{E}$, we can infer that $E(\lambda, S)$ is T -invariant for all $T \in \mathcal{E}$.

Then we can apply the induction hypothesis to $E(\lambda, S)$ and infer that there exists a vector in $E(\lambda, S)$ that is an eigenvector for every element of $\mathcal{E}$.

(b) We will use induction on $\dim V$. The case $\dim V = 1$ is trivial. Assume that the statement holds for any complex vector space with dimension less than n for some $n \geq 2$.

From (a), we can pick a vector v that is an eigenvector for every element of $\mathcal{E}$. Denote $U = \text{span}\{v\}$, and select a subspace W of V such that $V = U \oplus W$. Denote P as the projection from V to W along U . For any $T \in \mathcal{E}$, define $f(T) \in \mathcal{L}(W)$ by $f(T)(w) = P(Tw)$ for all $w \in W$. Then for any $S, T \in \mathcal{E}$, we have

$$f(S)f(T)(w) = P(SP(Tw)) = P(STw) = P(TSw) = P(TP(Sw)) = f(T)f(S)(w).$$

Therefore, $f(S)$ and $f(T)$ commute for all $S, T \in \mathcal{E}$. Since $\dim W < \dim V$, we can apply the induction hypothesis to W and infer that there exists a basis $\mathcal{B}'$ of W with respect to which every element of $f(\mathcal{E})$ has an upper-triangular matrix.

Consider the basis $\mathcal{B} = \{v\} \cup \mathcal{B}'$ of V . Then it's easy to check that for all $T \in \mathcal{E}$, we have

$$\mathcal{M}(T, \mathcal{B}) = \begin{pmatrix} \lambda & * \\ 0 & \mathcal{M}(f(T), \mathcal{B}') \end{pmatrix},$$

where $Tv = \lambda v$ for all $v \in U$. Since $\mathcal{M}(f(T), \mathcal{B}')$ is upper triangular, we have $\mathcal{M}(T, \mathcal{B})$ is also upper triangular. $\square$